

HOEDSPRUIT, South Africa

WMO: 682910

Lat: 24.352S

Lon: 31.049E

Elev: 524

StdP: 95.19

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	9.1	-5.0	2.6	22.4	-2.3	3.3	20.8	9.1	18.6	8.3	18.8	1.1	240	0.339

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.4	36.2	20.4	34.8	20.3	33.3	20.3	24.5	29.7	23.7	29.4	23.1	28.7	3.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	19.0	26.8	22.2	18.0	25.7	21.6	17.3	25.3	77.1	29.6	73.8	29.6	71.4	28.7	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.0	6.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	2.9	25.3	2.5	1.1	1.1	26.0	-0.4	26.7	-1.8	27.3	-3.6	28.1	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.2	25.1	25.2	24.7	22.3	20.0	18.3	17.6	19.5	21.9	23.3	23.9	25.1
	DBStd	3.82	2.32	2.15	2.15	2.64	2.33	2.27	2.22	2.92	3.48	3.59	3.27	2.88
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	114	0	0	0	2	9	27	41	21	9	3	2	0
	CDD10.0	4462	469	427	457	370	308	251	234	294	357	411	416	468
	CDD18.3	1534	211	193	198	122	59	27	17	57	116	156	168	209
	CDH23.3	13061	1676	1466	1502	907	529	285	218	561	1129	1509	1463	1815
	CDH26.7	5199	633	553	605	337	158	59	42	192	497	700	647	776
(14) Wind	WSAvg	2.9	3.2	3.2	2.7	2.5	2.2	2.4	2.6	2.8	3.3	3.5	3.1	3.0
(15) Precipitation	PrecAvg	624	116	95	88	40	11	5	5	7	16	39	86	111
	PrecMax	1293	261	455	239	163	69	26	20	28	73	77	217	213
	PrecMin	301	29	20	30	3	1	0	0	0	1	8	19	33
	PrecStd	210	66	98	55	34	13	6	5	7	16	21	48	47
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	36.2	36.1	34.9	33.1	30.8	30.2	33.4	36.8	37.7	37.1	38.0
		MCWB	21.6	21.6	21.6	19.9	17.7	15.0	13.4	16.1	17.3	19.5	20.2	21.6
	2%	DB	33.3	33.7	34.0	32.3	30.1	28.4	27.9	31.1	34.1	35.1	35.0	35.0
		MCWB	22.2	21.8	22.9	20.1	16.6	14.5	14.0	15.6	17.2	19.2	20.0	22.0
	5%	DB	31.8	31.8	32.1	30.2	28.2	26.5	25.9	29.0	31.9	32.9	32.9	33.0
		MCWB	22.0	21.6	21.4	19.7	16.4	14.4	13.9	15.3	17.3	19.1	20.2	22.0
	10%	DB	30.1	30.1	30.1	28.2	26.2	24.8	24.0	26.7	29.4	30.6	30.7	31.0
		MCWB	21.8	21.4	21.3	19.4	16.2	14.3	13.8	14.8	17.0	18.9	20.3	21.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	25.2	25.0	23.4	20.3	17.7	17.5	19.1	20.6	22.9	24.3	25.3
		MCDB	30.2	28.3	31.7	30.5	27.2	25.0	23.2	26.7	29.5	30.2	30.0	29.9
	2%	WB	24.2	23.9	23.6	21.6	18.7	16.7	16.2	18.0	19.6	21.6	22.9	23.9
		MCDB	29.4	28.7	29.9	27.7	25.4	23.5	22.3	25.7	28.1	29.8	28.6	30.2
	5%	WB	23.5	23.0	22.9	20.7	18.0	16.0	15.4	17.1	18.7	20.7	22.1	23.1
		MCDB	28.7	28.0	28.5	26.9	24.7	22.5	21.5	23.9	27.1	28.5	28.0	29.5
	10%	WB	22.8	22.4	22.2	20.1	17.2	15.3	14.7	16.2	17.9	19.9	21.3	22.4
		MCDB	27.7	27.2	27.1	26.0	23.7	21.6	20.8	23.1	26.0	27.4	27.5	28.5
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.4	10.7	11.4	13.2	13.6	13.0	13.1	12.6	11.6	10.6	10.1
		MCDBR	12.3	12.3	14.2	15.1	16.7	16.6	16.5	17.2	17.2	16.0	15.3	13.7
		MCWBR	3.6	3.4	4.6	5.0	6.2	6.7	6.6	6.2	5.9	4.8	4.5	4.2
	5% WB	MCDWR	9.7	9.3	10.8	11.7	13.2	12.3	12.1	12.6	13.3	13.3	11.1	11.0
		MCWBR	4.0	3.8	4.1	4.3	5.4	5.5	5.9	5.2	5.4	4.4	4.2	4.5
(40) Clear-Sky Solar Irradiance	taub		0.409	0.401	0.396	0.382	0.349	0.332	0.334	0.391	0.450	0.460	0.426	0.422
	taud		2.374	2.397	2.402	2.398	2.438	2.461	2.446	2.279	2.143	2.168	2.301	2.330
	Ebn at Noon		936	929	905	872	860	857	869	849	840	864	914	925
	Edh at Noon		131	125	119	110	96	90	94	122	152	156	140	137
(44) All-Sky Solar Radiation	RadAvg		6.19	6.04	5.38	4.59	4.24	3.80	4.00	4.67	5.28	5.51	5.85	6.09
	RadStd		0.42	0.47	0.39	0.30	0.18	0.23	0.14	0.28	0.34	0.35	0.49	0.51

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page